TEEN TAAL

Setting the stage and introducing the instruments

How large can a set of integers be without containing three equally spaced elements? We begin with the history of this question, move it to a finite cyclic group, and introduce its characters and Fourier transform. We finish by applying the transform to the three-term progression count.

1.1 Notation: arithmetic progressions

Definition 1.1. For an integer $k \geq 2$, a *k-term arithmetic progression* (k-AP) in $\mathbb{Z}$ is a sequence

$$x,\ x+d,\ \ldots,\ x+(k-1)d.$$

It is *nontrivial* if $d \neq 0$. A set is *k-AP-free* if it contains no nontrivial k-AP.

For example, $3, 8, 13$ is a nontrivial 3-AP. For three integers $a < b < c$, equal spacing is equivalent to

$$a + c = 2b. \tag{1.1}$$

The additive equation, rather than just the picture of equal gaps, will be what makes Fourier analysis useful. Write $[N] = \{1, \ldots, N\}$.

1.2 History: from colouring to density

1.2.1 *Van der Waerden's theorem (1927)*

Theorem 1.2 (Van der Waerden). *For every pair of integers $q \geq 1$, $k \geq 2$, there exists $W = W(q,k)$ such that every colouring of $[W]$ with q colours contains a nontrivial monochromatic k-term arithmetic progression.*

Equivalently, every finite colouring of $\mathbb{N}$ has monochromatic arithmetic progressions of every finite length. The theorem guarantees a progression in *some* colour class; it does not say that every infinite class has one.

Audience question. Can you exhibit an infinite set with no nontrivial three-term arithmetic progression?

The set $B = \{1, 2, 4, 8, \ldots\}$ works. If $2^a < 2^b < 2^c$ were equally spaced, then dividing $2^a + 2^c = 2^{b+1}$ by 2^a would give

$$1 + 2^{c-a} = 2^{b+1-a},$$

which equates an odd integer with an even one. Thus B is 3-AP-free, and consequently k-AP-free for every $k \geq 3$. This is an example of a *set*, not a colouring in which every colour class avoids progressions.

1.2.2 How thin is this example?

Definition 1.3. For $A \subseteq \mathbb{N}$, its *upper density* is

$$\overline{d}(A) = \limsup_{N\to\infty} \frac{|A \cap [N]|}{N}.$$

If the ordinary limit exists, it is the *natural density* $d(A)$.

For the powers of two,

$$|B \cap [N]| = \lfloor \log_2 N \rfloor + 1, \qquad \frac{|B \cap [N]|}{N} \longrightarrow 0.$$

Hence $d(B) = \overline{d}(B) = 0$. More generally, upper density zero implies that natural density exists and is zero: the ratios are nonnegative, so their liminf and limsup must both be zero.

The question is whether every progression-free set must be this thin in the sense of having density zero. For a finite formulation, define

$$r_k(N) = \max\{|A| : A \subseteq [N],\ A \text{ is } k\text{-AP-free}\}.$$

For each fixed $k \geq 3$, the density question asks whether $r_k(N)/N \to 0$.

1.2.3 Erdős–Turán, Szemerédi, and the reciprocal question

Erdős and Turán proposed the density conjecture in 1936. Szemerédi's 1975 theorem settled it for every fixed progression length:

Theorem 1.4 (Szemerédi)**.** *For every integer $k \geq 3$ and every $\delta > 0$, there exists $N_0(k, \delta)$ such that any $A \subseteq [N]$ with $N \geq N_0(k, \delta)$ and $|A| \geq \delta N$ contains a nontrivial k-term arithmetic progression.*

Thus $r_k(N) = o(N)$. In particular, an infinite set of positive upper density contains progressions of every finite length: choose arbitrarily large N for which its density in $[N]$ exceeds a fixed positive number.

The primes are another useful example of a zero-density set. They are not progression-free: Green and Tao proved that the primes contain arbitrarily long arithmetic progressions [4]. Their 2004 preprint appeared in print in 2008. Szemerédi's theorem by itself cannot yield this conclusion, because its hypothesis is positive density among the integers.

Erdős also asked a stronger question:

$$\sum_{a\in A} \frac{1}{a} = \infty \quad \Longrightarrow \quad A \text{ contains arithmetic progressions of every finite length?}$$

The three-term version is recorded in his seminar of 21 May 1973 [5]; the full version is documented in the mid-1970s [6]. Positive upper density implies divergence of the reciprocal sum, but the converse fails: there are zero-density sets with divergent reciprocal sum. For example, $a_n = \lceil n \log(n+1) \rceil$ is strictly increasing, $a_n/n \to \infty$, and $\sum_n 1/a_n$ diverges by comparison with $\sum_{n\geq 2} 1/(n \log n)$. We will not pursue reciprocal-sum estimates in this course.

2026 update. OpenAI's release of 6 October 2026 includes a preprint, *Quasipolynomial Bounds for Arithmetic Progressions*, announcing a proof of the full reciprocal-sum conjecture [9, 10].

1.2.4 Roth's first breakthrough (1953)

The first major density result was the case $k = 3$, proved by Roth in 1953 [1].

Theorem 1.5 (Roth). *For every $\delta > 0$, there exists $N_0(\delta)$ such that every $A \subseteq [N]$ with $N \geq N_0(\delta)$ and $|A| \geq \delta N$ contains a nontrivial three-term arithmetic progression. Equivalently, $r_3(N) = o(N)$.*

Roth was 27 when his paper appeared in 1953; he received the Fields Medal in 1958 [8]. Our aim is to understand the Fourier-analytic route to Theorem 1.5. Three lectures, three taals, one theorem.

1.3 Preparing the stage

1.3.1 Step I: reduction modulo $m = 2N + 1$

We wish to use periodic waves, so we place $[N]$ inside the cyclic group

$$G = \mathbb{Z}/m\mathbb{Z}, \qquad m = 2N + 1.$$

Let $\pi : \mathbb{Z} \to G$ be reduction modulo m, and let $\iota = \pi|_{[N]}$. Thus $\iota(x) = \overline{x}$, and $\iota : [N] \hookrightarrow G$ is injective. Keep the integer sets and their images distinct by writing

$$S = \iota([N]) \subseteq G, \qquad B = \iota(A) \subseteq S.$$

We have $|S| = N$ and $|B| = |A|$. More importantly, reduction preserves exactly the three-term progression equation.

Lemma 1.6 (No spurious three-term progressions). *For $x, y, z \in [N]$,*

$$\overline{x} + \overline{z} = 2\overline{y} \text{ in } G \quad \Longleftrightarrow \quad x + z = 2y \text{ in } \mathbb{Z}.$$

Consequently, $A \subseteq [N]$ is 3-AP-free in the integers if and only if $B = \iota(A)$ is 3-AP-free in G.

Proof. The forward congruence says that $m \mid x + z - 2y$. But

$$|x + z - 2y| \leq 2N - 2 < m,$$

so this multiple of m must be zero. The converse is immediate by reduction modulo m. Distinct integers in $[N]$ remain distinct modulo m, so nontriviality is preserved as well. □

In an odd-order cyclic group, a progression $x, x + d, x + 2d$ with $d \neq 0$ has three distinct terms: equality of the endpoints would give $2d = 0$, and multiplication by 2 is invertible modulo m.

1.3.2 Step II: indicator functions and progression counts

Definition 1.7. For $E \subseteq G$, its *indicator function* on G is

$$\mathbf{1}_E : G \to \{0, 1\}, \qquad \mathbf{1}_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E. \end{cases}$$

For any function on G, normalized averaging means $\mathbb{E}_{x \in G} f(x) = m^{-1} \sum_{x \in G} f(x)$.

Throughout the Fourier calculations, $A \subseteq [N]$ remains a set of integers, while $B \subseteq S \subseteq G$ denotes its image. In particular, $\mathbf{1}_B$ and $\mathbf{1}_S$ are functions on G. There are two densities, with the same cardinality in the numerator:

$$\delta = \frac{|A|}{N}, \qquad \alpha = \frac{|B|}{m} = \frac{N}{m}\delta. \tag{1.2}$$

Here δ is the actual density of the chosen A in $[N]$ and depends on A, N; α is the mean of $\mathbf{1}_B$ on G. Roth's theorem fixes a positive lower bound for the actual density as N varies.

Define the trilinear progression-counting functional

$$\Lambda_3(f_1, f_2, f_3) = \mathbb{E}_{x,d \in G} f_1(x) f_2(x+d) f_3(x+2d). \tag{1.3}$$

The product of three indicators equals one precisely when all three entries belong to B. Thus

$$m^2 \Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) = \#\{(x,d) \in G^2 : x, x+d, x+2d \in B\}.$$

There are exactly $|B| = |A|$ trivial contributions, namely $(x, 0)$ with $x \in B$. All other contributions are nonnegative. By Lemma 1.6,

$$A \text{ is 3-AP-free} \iff \Lambda_3(\mathbf{1}_B, \mathbf{1}_B, \mathbf{1}_B) = \frac{|B|}{m^2} = \frac{\alpha}{m}. \tag{1.4}$$

We have replaced a question about a set by an identity involving a function.

Remark 1.8 (The whole interval). For $A = [N]$, we have $B = S = \{\overline{n} : n \in [N]\} \subsetneq G$. Then

$$\begin{aligned} m^2 \Lambda_3(\mathbf{1}_S, \mathbf{1}_S, \mathbf{1}_S) &= \#\{(x,d) \in G^2 : x, x+d, x+2d \in S\} \\ &= \#\{(x,y,z) \in S^3 : x + z = 2y \text{ in } G\} \\ &= \#\{(a,b,c) \in [N]^3 : a + c = 2b \text{ in } \mathbb{Z}\} \\ &= \left\lceil \frac{N}{2} \right\rceil^2 + \left\lfloor \frac{N}{2} \right\rfloor^2 \geq \frac{N^2}{2}. \end{aligned} \tag{1.5}$$

The map $(x,d) \mapsto (x, x+d, x+2d)$ is a bijection onto the displayed triples in S^3, with inverse $(x,y,z) \mapsto (x, y-x)$. The equality of the two triple counts follows from Lemma 1.6. Finally, a, c must have the same parity, and each such ordered pair determines $b = (a+c)/2 \in [N]$. This count includes the N constant progressions and both orientations of every nonconstant progression.

Set $Q_N = m^2 \Lambda_3(\mathbf{1}_S, \mathbf{1}_S, \mathbf{1}_S)$ for use below.

1.4 Introducing the instruments

1.4.1 Instrument I: characters

Translation is a basic operation on G. For $d \in G$, put

$$(T_d f)(x) = f(x+d).$$

Which functions change only by a scalar factor when translated?

Definition 1.9. A *character* of G is a group homomorphism $\chi : G \to \mathbb{C}^\times$, where G is additive and $\mathbb{C}^\times$ is multiplicative. Thus $\chi(x+y) = \chi(x)\chi(y)$.

Every character takes values on the unit circle. Indeed, $\chi(x)^m = \chi(mx) = \chi(0) = 1$. For an integer t, write $e_m(t) = \exp(2\pi i t/m)$.

Proposition 1.10 (All characters of the cyclic group)**.** *The characters of G are precisely*

$$\chi_r(x) = e_m(rx), \qquad r \in G.$$

Moreover, $T_d\chi_r = e_m(rd)\chi_r$ for every $d \in G$.

Proof. The expression is independent of the chosen representatives modulo m, and $e_m(r(x+y)) = e_m(rx)e_m(ry)$, so each χ_r is a character. Conversely, a character is determined by $\chi(1)$. Since $\chi(1)^m = 1$, it equals $e_m(r)$ for a unique $r \in G$. Then $\chi(x) = \chi(1)^x = e_m(rx)$. Finally, $\chi_r(x+d) = e_m(rd)\chi_r(x)$. □

The characters are therefore simultaneous eigenfunctions of all translation operators. They are the basic waves adapted to the cyclic stage.

Lemma 1.11 (Orthogonality)**.** *For $t \in G$,*

$$\mathbb{E}_{x\in G} e_m(tx) = \begin{cases} 1, & t = 0, \\ 0, & t \neq 0. \end{cases}$$

In particular, $\mathbb{E}_x \chi_r(x)\overline{\chi_s(x)} = \mathbf{1}_{r=s}$.

Proof. For $t = 0$, every summand is one. Otherwise, let $z = e_m(t)$. Then $z \neq 1$ and $z^m = 1$, whence

$$\sum_{x=0}^{m-1} z^x = \frac{1 - z^m}{1 - z} = 0.$$

The last assertion follows by applying the first to $t = r - s$. □

Equip functions on G with the inner product

$$\langle f, g\rangle = \mathbb{E}_x f(x)\overline{g(x)}, \qquad \|f\|_2^2 = \mathbb{E}_x |f(x)|^2.$$

There are m orthonormal characters in the m-dimensional vector space of functions on G. They form an orthonormal basis.

1.4.2 Instrument II: Fourier methods

Definition 1.12 (Fourier transform)**.** For $f : G \to \mathbb{C}$, its Fourier coefficient at frequency r is

$$\widehat{f}(r) = \langle f, \chi_r\rangle = \frac{1}{m}\sum_{x\in G} f(x)e_m(-rx). \tag{1.6}$$

1.4.3 Application of the Fourier transform: counting progressions

Orthogonality gives the identity

$$\mathbf{1}_{\{0\}}(v) = \frac{1}{m}\sum_{r\in G} e_m(rv) \qquad (v \in G).$$

The map $(x,d) \mapsto (x, x+d, x+2d)$ is a bijection from G^2 onto the triples $(x,y,z) \in G^3$ satisfying $x+z = 2y$. Consequently, for functions $f_1, f_2, f_3 : G \to \mathbb{C}$,

$$\begin{aligned}\Lambda_3(f_1,f_2,f_3) &= \frac{1}{m^2}\sum_{\substack{x,y,z\in G\\ x+z=2y}} f_1(x)f_2(y)f_3(z)\\ &= \frac{1}{m^3}\sum_{r\in G}\left(\sum_{x\in G} f_1(x)e_m(rx)\right)\left(\sum_{y\in G} f_2(y)e_m(-2ry)\right)\left(\sum_{z\in G} f_3(z)e_m(rz)\right).\end{aligned}$$

Replace r by $-r$ in the final sum. The definition of the Fourier transform then gives

$$\boxed{\Lambda_3(f_1,f_2,f_3) = \sum_{r\in G}\widehat{f_1}(r)\widehat{f_2}(-2r)\widehat{f_3}(r).} \tag{1.7}$$

In particular, for an integer set $A \subseteq [N]$ and its image $B = \pi(A) \subseteq G$, this identity reads

$$\Lambda_3(\mathbf{1}_B,\mathbf{1}_B,\mathbf{1}_B) = \sum_{r\in G}\widehat{\mathbf{1}_B}(r)^2\widehat{\mathbf{1}_B}(-2r).$$

If A is 3-AP-free, the left-hand side equals $|A|/m^2$ by (1.4). Thus an integer problem has become an exact identity among Fourier coefficients on G.

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